

Jeremy Dao

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Education

- **Oregon State University** **Corvallis, OR**
PhD in Artificial Intelligence, Expected September 2025 *Sept. 2021 - Present*
Advisor: Alan Fern
- **Oregon State University** **Corvallis, OR**
M.S in Robotics *Sept. 2018 - Sept. 2021*
Thesis: Practical Reinforcement Learning for Bipedal Locomotion
Advisor: Alan Fern
- **University of Washington** **Seattle, WA**
B.S in Applied and Computational Math Sciences: Discrete Math and Algorithms, Minor in Physics, Math, and Music *Sept. 2014 - Sept. 2018*

Professional Experience

- **Oregon State University: Dynamic Robotics and Artificial Intelligence Laboratory** **Corvallis, OR**
Graduate Research Assistant - Supervisor: Alan Fern *Sept. 2018 - Present*
 - Researching sim-to-real methods for allowing consistent transfer of simulation trained control policies to real hardware.
 - Used reinforcement learning methods to create control policies for the bipedal robots Cassie and Digit that improve robustness and allow for new locomotion and manipulation capabilities.
 - Created infrastructure and wrapper code to interface with dynamic simulation libraries.
- **RAI Institute** **Cambridge, MA**
Research Intern - Supervisor: Farbod Farshidian *Oct. 2024 - Apr. 2025*
 - Investigated sim-to-real reinforcement learning techniques for quadruped and humanoid robots.
 - Fixed and refined existing model identification process code for more general use across different robot morphologies.
- **Agility Robotics** **Tangent, OR**
Software Engineer Intern - Supervisor: Taylor Apgar *Jun. 2022 - Dec. 2022*
 - Investigated sim-to-real reinforcement learning for locomotion control of the humanoid robot Digit.
 - Improved reinforcement learning code to speed up simulation throughput and training time.
- **GridRaster** **Palo Alto, CA**
Software Engineer Intern - Supervisor: Bhaskar Banerjee *Jun. 2019 - Sept. 2019*
 - Used off-the-shelf algorithms to train point cloud semantic segmentation networks for labeling 3D point cloud data.

Professional Experience (continued)

- **University of Washington: Movement Control Laboratory** **Seattle, WA**
Undergraduate Researcher - Supervisor: Kendall Lowrey *Oct. 2015 - May 2018*
and Emo Todorov
 - Worked with doctoral researchers to make Unscented Kalman Filter for a humanoid robot.
 - Contributed to policy and reward function testing of reinforcement learned policies for Phantom robots.

Research Publications

Journal Articles

- 1 K. Green, Y. Godse, **J. Dao**, R. L. Hatton, A. Fern, and J. Hurst, "Learning spring mass locomotion: Guiding policies with a reduced-order model," *IEEE Robotics and Automation Letters*, vol. 6, pp. 3926–3932, 2 2021.

Conference Proceedings

- 1 **J. Dao**, H. Duan, and A. Fern, "Sim-to-real learning for humanoid box loco-manipulation," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, May 2024, pp. 16 930–16 936.
- 2 H. Duan, B. Pandit, M. S. Gadde, B. V. Marum, **J. Dao**, C. Kim, and A. Fern, "Learning vision-based bipedal locomotion for challenging terrain," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, May 2024, pp. 56–62.
- 3 B. van Marum, A. Shrestha, H. Duan, P. Dugar, **J. Dao**, and A. Fern, "Revisiting reward design and evaluation for robust humanoid standing and walking," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, Oct. 2024, pp. 11 256–11 263.
- 4 D. Crowley, **J. Dao**, H. Duan, K. Green, J. Hurst, and A. Fern, "Optimizing bipedal locomotion for the 100m dash with comparison to human running," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, May 2023, pp. 12 205–12 211.
- 5 R. Batke, F. Yu, **J. Dao**, J. Hurst, R. L. Hatton, A. Fern, and K. Green, "Optimizing bipedal maneuvers of single rigid-body models for reinforcement learning," in *2022 IEEE-RAS 21st International Conference on Humanoid Robots (Humanoids)*, IEEE, Nov. 2022, pp. 714–721.
- 6 **J. Dao**, K. Green, H. Duan, A. Fern, and J. Hurst, "Sim-to-real learning for bipedal locomotion under unsensed dynamic loads," in *2022 International Conference on Robotics and Automation (ICRA)*, IEEE, May 2022, pp. 10 449–10 455.
- 7 H. Duan, A. Malik, **J. Dao**, A. Saxena, K. Green, J. Siekmann, A. Fern, and J. Hurst, "Sim-to-real learning of footstep-constrained bipedal dynamic walking," in *2022 International Conference on Robotics and Automation (ICRA)*, IEEE, May 2022, pp. 10 428–10 434.
- 8 H. Duan, A. Malik, M. S. Gadde, **J. Dao**, A. Fern, and J. Hurst, "Learning dynamic bipedal walking across stepping stones," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, Oct. 2022, pp. 6746–6752.
- 9 F. Yu, R. Batke, **J. Dao**, J. Hurst, K. Green, and A. Fern, "Dynamic bipedal turning through sim-to-real reinforcement learning," in *2022 IEEE-RAS 21st International Conference on Humanoid Robots (Humanoids)*, IEEE, Nov. 2022, pp. 903–910.

- 10 H. Duan, **J. Dao**, K. Green, T. Apgar, A. Fern, and J. Hurst, "Learning task space actions for bipedal locomotion," in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, Finalist for Best Paper Award in Cognitive Robotics, IEEE, May 2021, pp. 1276–1282.
- 11 J. Siekmann, S. Valluri, **J. Dao**, F. Bermillo, H. Duan, A. Fern, and J. Hurst, "Learning memory-based control for human-scale bipedal locomotion," in *Robotics: Science and Systems XVI*, Robotics: Science and Systems Foundation, Jul. 2020.
- 12 Z. Xie, P. Clary, **J. Dao**, P. Morais, J. Hurst, and M. V. D. Panne, "Learning locomotion skills for cassie: Iterative design and sim-to-real," in *Conference on Robotic Learning (CoRL)*, IEEE, 2019.
- 13 K. Lowrey, S. Koley, **J. Dao**, A. Rajeswaran, and E. Todorov, "Reinforcement learning for non-prehensile dynamic manipulation: Transfer from simulation to physical systems," in *IEEE International Conference on Simulation, 2018 Modeling and Programming for Autonomous Robots (SIMPAN)*, IEEE, 2018.
- 14 K. Lowrey, **J. Dao**, and E. Todorov, "Real-time state estimation with whole-body multi-contact dynamics: A modified ukf approach," in *2016 IEEE-RAS 16th International Conference on Humanoid Robots (Humanoids)*, IEEE, Nov. 2016, pp. 1225–1232.

Service

Mentoring

- Zachary Watts, Oregon State University Undergraduate Fall 2023 - Present
- Fangzhou Yu, Oregon State University MS Student Fall 2021 - Fall 2023
Recruited to RAI Institute
- Jonah Siekmann, Oregon State University Undergraduate Winter 2018 - Spring 2021
and MS Student
Recruited to Agility Robotics
- Yesh Godse, Oregon State University Undergraduate Winter 2018 - Spring 2021
Recruited to Agility Robotics

Peer Reviewer

- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Robotics (T-RO)
- IEEE Transactions on Automation Science and Engineering (T-ASE)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE International Conference on Intelligent Robots and Systems (IROS)
- Conference on Robot Learning (CoRL)
- Robotics: Science and Systems (RSS)

Awards and Honors

- Phi Beta Kappa Honor Society. 2018
- University of Washington Dean's List. 2014-2018
- Mary Gates Undergraduate Research Scholarship. 2016

Technical Skills

Skills	• Machine Learning, Reinforcement Learning, Search/Planning algorithms, Numerical Optimization
Coding Languages	• Python, C, C++, Java, Julia, Matlab
Software and Tools	• PyTorch, Mujoco, IsaacLab, Flux, Github, Jira, Agile, Kanban, L^AT_EX